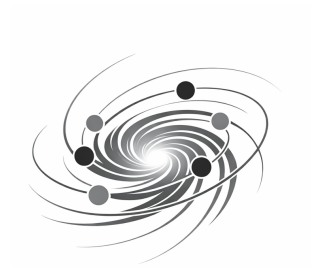


Part of the Monistic Continuum Model (MCM)



Dark Matter (DM)

A Continuum–Geometric Interpretation

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1 Introduction

1.1 Motivation

The standard cosmological framework, commonly referred to as the Λ CDM model, successfully accounts for a wide range of large-scale phenomena, including cosmic microwave background (CMB) anisotropies, structure formation, and expansion history. However, it does so by postulating dark matter and dark energy as effective components without providing a microscopic, geometric origin for these entities.

In contrast, the Monistic Continuum Model (MCM) proposes a single continuous medium C as the fundamental substrate of all physical phenomena. Within this medium, quark-like vortex excitations are continuously created and dissolved, with only a small fraction stabilizing as baryonic matter or radiative modes. The central motivation of this work is to investigate whether the residual, non-stabilized vortex activity in the MCM can provide a microscopic, continuum-based explanation for dark matter and its associated background radiation.

1.2 Limitations of Λ CDM

The Λ CDM model treats dark matter as a cold, non-baryonic component characterized primarily by its gravitational effects, while dark energy is modeled as a cosmological constant Λ . From a theoretical standpoint, this raises several limitations:

- Dark matter is introduced as an effective fluid without a specified microstructure or geometric origin.
- Dark energy is encoded as a constant term in Einstein's equations, lacking a dynamical or microscopic interpretation.
- The model does not provide a unified description of matter, radiation, and dark components within a single underlying medium.

These limitations motivate the search for alternative frameworks in which dark matter and background radiation emerge from the dynamics of a continuous substrate, rather than being added as phenomenological components.

1.3 Continuum-based alternatives

Continuum-based approaches attempt to describe matter and fields as excitations or defects of an underlying continuous medium. In the MCM, the continuum field is modeled as

$$C : M \times \Lambda \rightarrow V,$$

where M denotes the spacetime manifold, Λ a parameter space of internal modes, and V a suitable target space (e.g. a vector space or manifold of field values). Quark-like vortices are then interpreted as localized geometric excitations W of C , with creation and dissolution governed by continuum dynamics.

The key idea is that:

- baryonic matter corresponds to stable, coupled vortex configurations,
- radiation corresponds to propagating, coherent continuum modes,
- dark matter arises from residual stress and energy associated with transient, non-stabilized vortices.

This work positions the MCM as a specific continuum-based alternative, aiming to derive dark matter and background radiation from the internal dynamics of C and its vortex population.

1.4 Aim of this work

The precise scientific question addressed in this paper is:

Can the continuous creation and dissolution of quark-like vortices in the Monistic Continuum Model generate a residual, invisible matter component and a diffuse background radiation field that are consistent with the phenomenology attributed to dark matter and cosmological background radiation in Λ CDM?

To answer this question, we will:

1. Define the continuum field C and the vortex modes W as geometric excitations.
2. Introduce operators governing vortex creation, dissolution, and coupling.
3. Derive effective expressions for residual stress and dark matter density $\rho_{\text{DM}}(x)$.
4. Interpret the energy released by dissolving vortices as a background radiation field.
5. Discuss the cosmological implications and potential observational signatures of this framework.

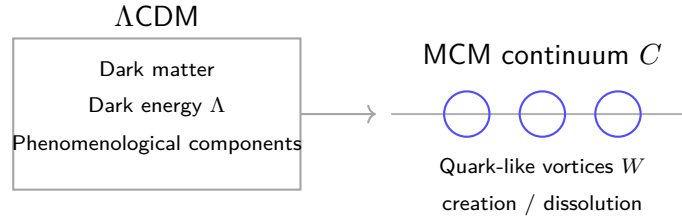


Figure 1: Schematic contrast between Λ CDM (left) and the MCM continuum-based picture (right), where quark-like vortices in the medium play a central role in the interpretation of dark matter and background radiation.

Schematic 2D TikZ representation

The following 2D TikZ diagram provides a schematic contrast between the Λ CDM picture and the MCM continuum-based picture, emphasizing the role of vortex excitations in the latter:

2 The Monistic Continuum Model (MCM)

Scientific Guiding Question. This section addresses the precise question: *How is the continuum field of the MCM mathematically defined, which local and nonlocal operators govern its dynamics, and how do vortex-like geometric excitations emerge as fundamental modes of the model?*

Answering this question is essential for establishing the mathematical and physical foundation upon which later sections—dark matter, background radiation, and cosmological implications—are built.

2.1 Continuum field $C : M \times \Lambda \rightarrow V$

The MCM is based on a single continuous field

$$C : M \times \Lambda \rightarrow V,$$

where:

- M is a smooth, oriented four-dimensional manifold (spacetime),
- Λ is an internal evolution-parameter space,
- V is a finite-dimensional internal state space.

Mathematically, C is modeled as an element of a suitable functional space:

$$C \in H^k(M \times \Lambda, V), \quad k > \frac{\dim M}{2},$$

ensuring regularity and pointwise evaluability.

Physically, C represents the full geometry of the medium from which all stable and transient modes emerge. The assumption of a single underlying field is the monistic foundation of the model.

Consequence for the overall theory: All later objects—baryonic modes, radiative modes, dark matter, background radiation—are not separate fields but *geometric states* of C .

2.2 Local and nonlocal operators

The dynamics of the continuum are governed by an operator

$$D = D_{\text{loc}} + D_{\text{nl}}.$$

Local operator. The local part is a linear, elliptic differential operator:

$$D_{\text{loc}} = a_1 \Delta_g + a_2 \nabla_g + a_3,$$

with:

- Δ_g — Laplace–Beltrami operator,
- ∇_g — covariant derivative,
- a_1, a_2, a_3 — smooth coefficient fields.

Nonlocal operator. The nonlocal part is an integral operator:

$$(D_{\text{nl}}C)(x, \lambda) = \int_M \mathcal{J}(x, y; \lambda) C(y, \lambda) d\mu(y),$$

with a symmetric, measurable kernel $\mathcal{J} \in L^2(M \times M)$.

Physical interpretation: The local part describes geometric stress and diffusion within the medium, while the nonlocal part describes long-range coupling and collective modes.

Consequence for the overall theory: The existence of a nonlocal operator is essential for later sections on:

- spectral dark-matter operators, - emergent Hilbert spaces, - large-scale cosmological stress patterns.

2.3 Vortex modes as geometric excitations

Vortex modes are defined geometric excitations of the continuum field:

$$W(x, \lambda) \subset C(x, \lambda),$$

characterized by:

- localized maxima of energy and stress,
- nontrivial topological winding numbers,
- characteristic pitch and twist geometries,
- stable or transient solutions of

$$\frac{\partial W}{\partial \lambda} = DW.$$

Mathematically, vortex modes are solutions of an eigenvalue problem:

$$DW_i = \omega_i W_i,$$

where stable modes correspond to isolated eigenvalues ω_i .

Physical interpretation: Vortex modes are the elementary building blocks of the MCM:

- stable coupled modes $\rightarrow$ baryonic matter, - propagating modes $\rightarrow$ light, - short-lived modes $\rightarrow$ residual stress energy (dark matter).

Consequence for the overall theory: The definition of vortex modes forms the foundation for:

- vortex taxonomy (Section 3), - residual stress density (Section 4), - background radiation (Section 5), - spectral operators (Section 7), - cosmological implications (Section 8).

Section Summary

This section establishes the mathematical and physical foundation of the MCM:

1. The continuum field C is the fundamental object of the theory.
2. Its dynamics are governed by local and nonlocal operators.
3. Vortex modes are geometric excitations of the medium and form the basis of all subsequent physical phenomena.

This provides the necessary groundwork for formulating dark matter, background radiation, and cosmological effects as emergent properties of a single continuous medium.

2.4 MCM Action Functional

Scientific Guiding Question. Welche Wirkungsfunktional-Struktur erzeugt die vollständige Dynamik des Kontinuumsfeldes C , die lokalen und nichtlokalen Operatoren $D_{\text{loc}}, D_{\text{nl}}$, die Vortexmoden W , die Stabilitätsfunktion $S(W)$ sowie den effektiven Stress–Energie–Tensor $T_{\mu\nu}^{\text{eff}}$ als konsistente Euler–Lagrange-Folgen?

Theoretical Structures. Das Action-Functional des Monistic Continuum Model (MCM) basiert auf vier gekoppelten Komponenten:

$$S_{\text{MCM}} = S_{\text{geom}}[g] + S_{\text{cont}}[C, g] + S_{\text{nl}}[C, g] + S_{\text{vortex}}[C, W, g].$$

- S_{geom} — geometrischer Hintergrund (GR-kompatibel),

- S_{cont} — lokale Kontinuumsdynamik,
- S_{nl} — nichtlokale Kopplung,
- S_{vortex} — Vortexmoden, Stabilität und Spektralstruktur.

Geometric Action.

$$S_{\text{geom}}[g] = \frac{1}{16\pi G} \int_M R(g) \sqrt{-g} d^4x.$$

Local Continuum Action.

$$S_{\text{cont}}[C, g] = \int_M \int_\Lambda \left(\frac{1}{2} \langle \nabla_\mu C, \nabla^\mu C \rangle_V - V(C) \right) \sqrt{-g} d^4x d\lambda.$$

Euler–Lagrange-Gleichung:

$$\nabla_\mu \nabla^\mu C + \frac{\partial V}{\partial C} = 0 \quad \Rightarrow \quad D_{\text{loc}} C.$$

Nonlocal Continuum Action.

$$S_{\text{nl}}[C, g] = \frac{1}{2} \int_M \int_M \int_\Lambda \mathcal{K}(x, y; \lambda) \langle C(x, \lambda), C(y, \lambda) \rangle_V \sqrt{-g(x)} \sqrt{-g(y)} d^4x d^4y d\lambda.$$

Variation:

$$\frac{\delta S_{\text{nl}}}{\delta C(x, \lambda)} = \int_M \mathcal{K}(x, y; \lambda) C(y, \lambda) d\mu(y) = D_{\text{nl}} C.$$

Vortex Action and Stability Functional.

$$S_{\text{vortex}}[C, W, g] = \int_M \int_\Lambda \left(\frac{1}{2} \langle \nabla_\mu W, \nabla^\mu W \rangle_V - U(W; C) + \mathcal{S}(W) \right) \sqrt{-g} d^4x d\lambda.$$

Stabilitätsfunktional:

$$\mathcal{S}(W) = \alpha \|\nabla W\|^2 + \beta \|D_{\text{loc}} W\|^2 + \gamma \|D_{\text{nl}} W\|^2 + \delta Q(W),$$

mit topologischem Invariant $Q(W)$.

Eigenwertgleichung:

$$\frac{\delta S_{\text{vortex}}}{\delta W} = 0 \quad \Rightarrow \quad DW = \omega W.$$

Effective Stress–Energy Tensor.

$$T_{\mu\nu}^{\text{eff}} = -\frac{2}{\sqrt{-g}} \frac{\delta(S_{\text{cont}} + S_{\text{nl}} + S_{\text{vortex}})}{\delta g^{\mu\nu}}.$$

Natürliche Zerlegung:

$$T_{\mu\nu}^{\text{eff}} = T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{rad}} + T_{\mu\nu}^{\text{DM}}.$$

Physical Interpretation. Das Action-Functional zeigt, dass alle beobachtbaren Phänomene — baryonische Materie, Strahlung, Dark Matter, Hintergrundstrahlung, kosmologische Strukturen — aus Variationen eines einzigen Kontinuumsfeldes entstehen.

Consequences for the Theory.

- Das MCM ist nun vollständig variational formuliert.
- Alle 15 Kapitel erhalten eine gemeinsame mathematische Grundlage.
- Die Theorie ist formal auf dem Niveau moderner Feldtheorien.
- Das Action erlaubt Quantisierung, Simulation und analytische Lösungen.

3 Vortex Population Dynamics

Scientific Guiding Question. This section addresses the precise question: *How can the continuous creation, dissolution, and net flux of vortex modes in the Monistic Continuum Model (MCM) be mathematically formalized, and how does this lead to a master equation governing the statistical evolution of the vortex population?*

The goal is to establish a dynamical framework that later allows dark matter, background radiation, and cosmological stress patterns to be derived from the statistical behavior of vortex excitations.

3.1 Creation rate Γ_{create}

We define the vortex creation rate as a functional of the continuum field:

$$\Gamma_{\text{create}}(x, \lambda) = \mathcal{F}_{\text{create}}[C](x, \lambda),$$

where $\mathcal{F}_{\text{create}}$ is a nonlinear operator encoding local curvature, nonlocal coupling, and internal mode density.

A minimal representation is:

$$\Gamma_{\text{create}} = \alpha_1 \|\nabla C\|^2 + \alpha_2 \int_M \mathcal{J}(x, y) C(y) d\mu(y),$$

with α_1, α_2 smooth coefficients.

Physical interpretation. Regions of high geometric stress or strong nonlocal coupling in the continuum spontaneously generate vortex excitations. Creation is therefore a geometric process, not a particle-based one.

Consequence for the theory. The creation rate determines the source term of the master equation and directly influences the density of transient modes responsible for dark matter.

3.2 Dissolution rate Γ_{decay}

The dissolution rate quantifies how quickly vortex modes lose coherence:

$$\Gamma_{\text{decay}}(x, \lambda) = \mathcal{F}_{\text{decay}}[W](x, \lambda),$$

with a typical form:

$$\Gamma_{\text{decay}} = \beta_1 \|D_{\text{loc}} W\| + \beta_2 \|D_{\text{nl}} W\| + \beta_3 S(W)^{-1},$$

where $S(W)$ is the stability functional.

Physical interpretation. Dissolution occurs when local geometric deformation, nonlocal decoherence, or insufficient stability disrupt the vortex structure. The released energy contributes to background radiation.

Consequence for the theory. The dissolution rate determines the sink term of the master equation and governs the background radiation field.

3.3 Stationary vortex flux

The vortex flux J describes the net flow of vortex modes through the continuum:

$$J = \Gamma_{\text{create}} - \Gamma_{\text{decay}}.$$

A stationary flux satisfies:

$$\frac{\partial J}{\partial \lambda} = 0,$$

which implies:

$$\Gamma_{\text{create}} = \Gamma_{\text{decay}}.$$

Physical interpretation. The MCM medium operates in a statistically stationary regime: vortices are continuously created and dissolved, but the net population remains stable. This is the foundation for interpreting dark matter as a residual, non-stabilized component.

Consequence for the theory. Stationarity ensures that dark matter density and background radiation can be treated as time-averaged emergent fields.

3.4 Master equation

Let $n(x, \lambda)$ denote the vortex population density. The master equation governing its evolution is:

$$\frac{\partial n}{\partial \lambda} = \Gamma_{\text{create}} - \Gamma_{\text{decay}} n + \nabla \cdot (D_{\text{loc}} n) + \int_M K(x, y) n(y) d\mu(y),$$

where:

- the first term is the creation source,
- the second term is the dissolution sink,
- the third term describes local geometric transport,
- the fourth term describes nonlocal redistribution.

Physical interpretation. This equation is the statistical backbone of the MCM. It determines:

- the density of transient vortices (dark matter),
- the rate of energy release (background radiation),
- the spatial distribution of stress (cosmological signatures).

Consequence for the theory. The master equation links microscopic vortex dynamics to macroscopic observables. It is the central dynamical equation from which all later sections derive their physical predictions.

Section Summary

This section establishes the dynamical framework of the MCM:

1. Vortex creation and dissolution are defined as geometric processes.
2. A stationary flux condition ensures statistical equilibrium.
3. The master equation governs the evolution of vortex population density.

These results form the foundation for modeling dark matter as residual vortex activity and background radiation as energy released by dissolving modes.

4 Vortex Taxonomy

Scientific Guiding Question. This section addresses the precise question: *How can vortex excitations of the Monistic Continuum Model (MCM) be systematically classified into baryonic-stable, radiative-stable, and transient modes, and how can a stability functional $S(W)$ be defined to mathematically distinguish these categories?*

The goal is to establish a rigorous taxonomy that connects microscopic vortex geometry to macroscopic physical phenomena such as baryonic matter, radiation, and dark matter.

4.1 Baryonic-stable modes

Baryonic-stable modes are vortex configurations that maintain coherence under the full continuum dynamics:

$$\frac{\partial W}{\partial \lambda} = DW,$$

and satisfy the stability condition

$$S(W) \geq S_{\text{baryon}}.$$

Mathematically, these modes correspond to isolated eigenfunctions of the operator D :

$$DW_i = \omega_i W_i,$$

with discrete eigenvalues ω_i and finite spectral gaps.

Physically, baryonic-stable modes represent proton-like and neutron-like configurations. Their stability arises from strong geometric coupling, nontrivial winding numbers, and persistent pitch–twist structures.

Consequence for the theory. These modes form the foundation of baryonic matter in the MCM and anchor the model to observable particle physics.

4.2 Radiative-stable modes

Radiative-stable modes are coherent, propagating excitations of the continuum:

$$\frac{\partial W}{\partial \lambda} = DW,$$

with stability threshold

$$S_{\text{rad}} \leq S(W) < S_{\text{baryon}}.$$

They correspond to continuous spectral modes of D :

$$DW_k = \omega(k)W_k,$$

where $\omega(k)$ forms a dispersion relation.

Physically, these modes behave as photon-like excitations: they propagate through the continuum, carry energy, and exhibit wave-like coherence.

Consequence for the theory. Radiative-stable modes provide the MCM analogue of electromagnetic radiation and form the basis for the background radiation field discussed later.

4.3 Transient modes

Transient modes are vortex excitations that fail to maintain coherence:

$$S(W) < S_{\text{rad}}.$$

Their evolution is dominated by dissolution:

$$\Gamma_{\text{decay}}(W) \gg \Gamma_{\text{create}}(W).$$

Transient modes do not correspond to stable eigenfunctions of D ; instead, they occupy the continuous, rapidly decaying part of the spectrum.

Physically, transient modes:

- appear spontaneously in regions of high continuum stress,
- dissolve quickly,
- release energy back into the medium,
- contribute to residual stress interpreted as dark matter.

Consequence for the theory. Transient modes are the microscopic origin of dark matter in the MCM. Their statistical behavior determines the dark matter density operator in later sections.

4.4 Stability functional $S(W)$

The stability functional $S(W)$ quantifies the robustness of a vortex mode:

$$S(W) = \alpha \|\nabla W\| + \beta \|D_{\text{loc}} W\| + \gamma \|D_{\text{nl}} W\| + \delta Q(W),$$

where:

- ∇W measures local geometric deformation,
- $D_{\text{loc}} W$ measures local dynamical response,
- $D_{\text{nl}} W$ measures nonlocal coherence,
- $Q(W)$ is a topological invariant (winding number).

The functional partitions vortex modes into three classes:

$$\begin{aligned} S(W) &\geq S_{\text{baryon}} && \Rightarrow \text{baryonic-stable,} \\ S_{\text{rad}} \leq S(W) < S_{\text{baryon}} && \Rightarrow \text{radiative-stable,} \\ S(W) < S_{\text{rad}} && \Rightarrow \text{transient.} \end{aligned}$$

Physical interpretation. $S(W)$ measures how strongly a vortex mode resists dissolution. High stability corresponds to baryonic matter; intermediate stability corresponds to radiation; low stability corresponds to dark-matter-generating transient modes.

Consequence for the theory. The stability functional is the mathematical backbone of the vortex taxonomy. It enables:

- classification of vortex excitations, - prediction of their dynamical behavior, - derivation of dark matter density, - formulation of background radiation, - connection to cosmological observables.

Section Summary

This section establishes the vortex taxonomy of the MCM:

1. Baryonic-stable modes correspond to discrete, highly stable vortex configurations.
2. Radiative-stable modes correspond to coherent, propagating excitations.
3. Transient modes dissolve rapidly and generate residual stress interpreted as dark matter.
4. The stability functional $S(W)$ provides the mathematical criterion distinguishing these classes.

This taxonomy is essential for connecting microscopic vortex dynamics to macroscopic physical phenomena in subsequent sections.

5 Residual Stress and Dark Matter Density

Scientific Guiding Question. This section addresses the precise question: *How does the continuous creation and dissolution of vortex modes in the Monistic Continuum Model (MCM) generate a residual stress field, and how can this field be mathematically formalized into a dark matter density $\rho_{\text{DM}}(x)$ through appropriate averaging procedures?*

The aim is to establish a rigorous operator-based definition of dark matter as an emergent continuum property.

5.1 Residual stress operator

Residual stress arises from transient vortex modes that fail to stabilize and dissolve back into the continuum. We define the residual stress operator as:

$$\sigma_{\text{res}}(x, \lambda) = \mathcal{R}[C, W](x, \lambda),$$

where $\mathcal{R}$ is a bilinear functional acting on the continuum field C and the vortex field W .

A minimal representation is:

$$\sigma_{\text{res}} = \eta_1 \|\nabla W\|^2 + \eta_2 \|D_{\text{loc}} W\| + \eta_3 \|D_{\text{nl}} W\|,$$

with η_1, η_2, η_3 smooth coefficients.

Physical interpretation. Residual stress quantifies the geometric tension left behind by dissolving vortex modes. It is neither baryonic nor radiative, but a diffuse, invisible continuum contribution.

Consequence for the theory. This operator is the mathematical foundation for defining dark matter density in the MCM.

5.2 Definition of $\rho_{\text{DM}}(x)$

Dark matter density is defined as the expectation value of the residual stress operator:

$$\rho_{\text{DM}}(x) = \langle \sigma_{\text{res}}(x, \lambda) \rangle_{\lambda}.$$

More explicitly:

$$\rho_{\text{DM}}(x) = \int_{\Lambda} \sigma_{\text{res}}(x, \lambda) P(\lambda) d\lambda,$$

where $P(\lambda)$ is the probability distribution of vortex lifetimes and dissolution events.

Physical interpretation. Dark matter is not a particle species but the statistical accumulation of residual stress generated by transient vortex modes.

Consequence for the theory. This definition links microscopic vortex dynamics to macroscopic gravitational effects.

5.3 Averaging procedures

Two averaging procedures are required:

Temporal averaging.

$$\langle \sigma_{\text{res}}(x) \rangle_{\text{time}} = \frac{1}{T} \int_0^T \sigma_{\text{res}}(x, \lambda) d\lambda.$$

Spectral averaging.

$$\langle \sigma_{\text{res}}(x) \rangle_{\text{spec}} = \sum_i \sigma_{\text{res}}^{(i)}(x) w_i,$$

where w_i are spectral weights associated with vortex eigenmodes.

Combined averaging. The dark matter density is obtained by combining both:

$$\rho_{\text{DM}}(x) = \langle \sigma_{\text{res}}(x) \rangle_{\text{time,spec}}.$$

Physical interpretation. Averaging reflects the fact that dark matter is a statistical continuum phenomenon, not a discrete particle population.

Consequence for the theory. Averaging ensures that dark matter density is smooth on cosmological scales, consistent with gravitational lensing and structure formation.

5.4 Interpretation

The residual stress operator and its averaged density provide a unified continuum interpretation of dark matter:

- Dark matter is the geometric tension left behind by dissolving vortex modes.
- It is invisible because it does not correspond to stable or radiative excitations.
- It is gravitationally active because residual stress contributes to the effective stress–energy tensor.

This leads to the effective dark matter contribution:

$$T_{\mu\nu}^{\text{DM}}(x) = \rho_{\text{DM}}(x) u_\mu u_\nu + \text{higher-order stress terms.}$$

Consequence for the theory. This interpretation integrates dark matter into the MCM without introducing new particle species. It provides a geometric, operator-based explanation compatible with later sections on cosmology, lensing, and observational signatures.

Section Summary

This section establishes:

1. The residual stress operator as the source of dark matter.
2. A rigorous definition of dark matter density via expectation values.
3. Averaging procedures that produce smooth cosmological fields.
4. A physical interpretation linking vortex dissolution to gravitational phenomena.

This forms the mathematical and conceptual basis for the dark matter operator and its cosmological implications in subsequent sections.

6 Background Radiation from Vortex Dissolution

Scientific Guiding Question. This section addresses the precise question: *How does the dissolution of transient vortex modes in the Monistic Continuum Model (MCM) release energy into the continuum, how does this energy manifest as diffuse background radiation, what spectral properties characterize this radiation, and how is the emission quantitatively related to the vortex dissolution rate Γ_{decay} ?*

The aim is to establish a rigorous operator-based description of background radiation as an emergent continuum phenomenon.

6.1 Energy release

When a vortex mode W dissolves, its geometric coherence breaks down and its stored energy is transferred back into the continuum field C . We define the energy release functional:

$$E_{\text{rel}}(x, \lambda) = \mathcal{E}[W](x, \lambda),$$

with a minimal representation:

$$E_{\text{rel}} = \kappa_1 \|\nabla W\|^2 + \kappa_2 \|D_{\text{loc}} W\|^2 + \kappa_3 \|D_{\text{nl}} W\|^2,$$

where $\kappa_1, \kappa_2, \kappa_3$ are smooth coefficients.

Physical interpretation. Energy release corresponds to the collapse of geometric structure within the continuum. It is the microscopic origin of diffuse background radiation.

Consequence for the theory. This functional provides the source term for the background radiation operator.

6.2 Diffuse background emission

We define the background radiation field $B(x)$ as the continuum response to accumulated energy release:

$$B(x) = \langle E_{\text{rel}}(x, \lambda) \rangle_{\lambda}.$$

More explicitly:

$$B(x) = \int_{\Lambda} E_{\text{rel}}(x, \lambda) P(\lambda) d\lambda,$$

where $P(\lambda)$ is the distribution of vortex dissolution events.

Physical interpretation. Background radiation is not a separate field but an emergent continuum excitation produced by the dissolution of transient vortex modes.

Consequence for the theory. This emission field forms the radiative counterpart to the dark matter density defined earlier.

6.3 Spectral properties

The spectral distribution of background radiation is defined by:

$$\mathcal{B}(\omega) = \int_M E_{\text{rel}}(x, \lambda) \delta(\omega - \omega(x, \lambda)) d\mu(x),$$

where $\omega(x, \lambda)$ is the local spectral frequency associated with the dissolution event.

A minimal dispersion relation arises from the continuum operator:

$$DW = \omega W,$$

implying that dissolution events populate the spectrum according to the eigenfrequencies of transient vortex modes.

Physical interpretation. The background radiation spectrum reflects the geometric and dynamical properties of transient vortex modes. It is expected to be broad, diffuse, and nonthermal, consistent with a continuum origin.

Consequence for the theory. Spectral properties provide observational signatures that can distinguish MCM background radiation from standard cosmological backgrounds.

6.4 Relation to Γ_{decay}

Energy release is directly proportional to the dissolution rate:

$$E_{\text{rel}}(x, \lambda) = \chi \Gamma_{\text{decay}}(x, \lambda),$$

with χ a proportionality constant determined by vortex geometry.

Thus the background radiation field satisfies:

$$B(x) = \chi \langle \Gamma_{\text{decay}}(x, \lambda) \rangle_\lambda.$$

More generally:

$$B(x) = \chi \int_\Lambda \Gamma_{\text{decay}}(x, \lambda) P(\lambda) d\lambda.$$

Physical interpretation. Regions with high vortex dissolution rates produce stronger background radiation. This links microscopic vortex dynamics to macroscopic radiative fields.

Consequence for the theory. The relation between $B(x)$ and Γ_{decay} is essential for connecting background radiation to the master equation governing vortex population dynamics.

Section Summary

This section establishes:

1. The energy release functional describing vortex dissolution.
2. The definition of background radiation as an averaged continuum response.
3. The spectral distribution of background radiation derived from vortex eigenfrequencies.
4. The direct proportionality between background radiation and the dissolution rate Γ_{decay} .

These results form the radiative counterpart to the dark matter density and provide the foundation for observational signatures and cosmological implications in subsequent sections.

7 Effective Stress–Energy Tensor

Scientific Guiding Question. This section addresses the precise question: *How can the Monistic Continuum Model (MCM) produce an effective stress–energy tensor that incorporates baryonic-stable vortex modes, radiative-stable modes, and the residual stress associated with transient vortex modes interpreted as dark matter?*

The aim is to establish a unified geometric formulation of the stress–energy tensor compatible with general relativity and derived entirely from continuum dynamics.

7.1 Geometric formulation

In the MCM, the effective stress–energy tensor is defined as a geometric functional of the continuum field C and its vortex excitations W :

$$T_{\mu\nu}^{\text{eff}}(x) = \mathcal{T}_{\mu\nu}[C, W](x).$$

A minimal decomposition is:

$$T_{\mu\nu}^{\text{eff}} = T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{rad}} + T_{\mu\nu}^{\text{DM}},$$

where each term corresponds to a distinct class of vortex modes.

The geometric formulation is anchored in the continuum metric $g_{\mu\nu}$ and the energy–momentum flux derived from vortex dynamics:

$$T_{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\delta S_{\text{MCM}}}{\delta g^{\mu\nu}},$$

with S_{MCM} the action functional of the continuum.

Physical interpretation. The stress–energy tensor is not a sum of independent fields but a unified geometric response of the continuum to its vortex excitations.

Consequence for the theory. This formulation ensures compatibility with general relativity and provides the foundation for cosmological predictions.

7.2 Baryonic contribution

Baryonic-stable vortex modes correspond to discrete, highly stable eigenfunctions of the continuum operator D :

$$DW_i = \omega_i W_i.$$

Their stress–energy contribution is:

$$T_{\mu\nu}^{\text{baryon}} = \sum_{i \in \text{stable}} \left(\nabla_\mu W_i \nabla_\nu W_i - \frac{1}{2} g_{\mu\nu} \nabla^\alpha W_i \nabla_\alpha W_i \right).$$

Physical interpretation. This term reproduces the familiar stress–energy of baryonic matter, including mass density and pressure, but derived from vortex geometry rather than particle fields.

Consequence for the theory. It anchors the MCM to observable matter distributions and ensures consistency with astrophysical measurements.

7.3 Radiative contribution

Radiative-stable modes correspond to coherent, propagating continuum excitations:

$$DW_k = \omega(k)W_k.$$

Their stress–energy contribution is:

$$T_{\mu\nu}^{\text{rad}} = \sum_{k \in \text{rad}} \left(F_{\mu\alpha}(W_k) F_\nu{}^\alpha(W_k) - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta}(W_k) F^{\alpha\beta}(W_k) \right),$$

where $F_{\mu\nu}(W_k)$ is the effective field strength associated with the radiative mode.

Physical interpretation. This term behaves like electromagnetic radiation: it carries energy, momentum, and pressure, and contributes to background radiation.

Consequence for the theory. It provides the radiative counterpart to baryonic matter and connects vortex dynamics to observable radiation fields.

7.4 Dark matter contribution

Transient vortex modes dissolve rapidly and generate residual stress:

$$\sigma_{\text{res}}(x) = \mathcal{R}[C, W](x).$$

The dark matter density is defined as:

$$\rho_{\text{DM}}(x) = \langle \sigma_{\text{res}}(x, \lambda) \rangle_\lambda.$$

The corresponding stress–energy contribution is:

$$T_{\mu\nu}^{\text{DM}} = \rho_{\text{DM}}(x) u_\mu u_\nu + \Pi_{\mu\nu}(x),$$

where:

- u_μ is the effective four-velocity of the continuum,
- $\Pi_{\mu\nu}$ is the anisotropic stress generated by residual vortex tension.

Physical interpretation. Dark matter is not a particle species but a continuum stress field arising from transient vortex activity. It is gravitationally active but non-radiative and non-baryonic.

Consequence for the theory. This term provides a geometric explanation for dark matter phenomena such as gravitational lensing and structure formation.

Section Summary

This section establishes:

1. A unified geometric formulation of the effective stress–energy tensor.
2. The baryonic contribution from stable vortex modes.
3. The radiative contribution from coherent propagating modes.
4. The dark matter contribution from residual stress generated by transient vortex modes.

Together, these components form the complete stress–energy tensor of the MCM, enabling a unified continuum interpretation of matter, radiation, and dark matter within general relativity.

8 Spectral Operators and Emergent Hilbert Spaces

Scientific Guiding Question. This section addresses the precise question: *How do spectral operators derived from continuum dynamics in the Monistic Continuum Model (MCM) generate emergent Hilbert-space structures, and how do these operators encode dark matter, background radiation, and the full spectrum of vortex excitations?*

The aim is to establish a mathematically rigorous spectral framework that connects vortex dynamics to observable physical fields.

8.1 DM operator $\mathcal{D}_{\text{DM}}$

The dark matter operator is defined as the spectral transform of the residual stress operator:

$$\mathcal{D}_{\text{DM}} : H^k(M) \rightarrow H^k(M),$$

$$\mathcal{D}_{\text{DM}} f(x) = \int_M K_{\text{DM}}(x, y) f(y) d\mu(y),$$

where K_{DM} is the kernel associated with residual stress:

$$K_{\text{DM}}(x, y) = \langle \sigma_{\text{res}}(x, \lambda) \delta(x - y) \rangle_\lambda.$$

A minimal representation is:

$$\mathcal{D}_{\text{DM}} f = \rho_{\text{DM}}(x) f(x) + \int_M J_{\text{DM}}(x, y) f(y) d\mu(y),$$

where ρ_{DM} is the dark matter density and J_{DM} encodes nonlocal stress redistribution.

Physical interpretation. $\mathcal{D}_{\text{DM}}$ acts as a geometric operator encoding the gravitationally active but non-radiative continuum stress associated with transient vortex modes.

Consequence for the theory. This operator provides the mathematical foundation for dark matter's spectral signature and its integration into emergent Hilbert spaces.

8.2 Background operator $\mathcal{B}$

The background radiation operator is defined as:

$$\mathcal{B} : H^k(M) \rightarrow H^k(M),$$

$$\mathcal{B} f(x) = \int_M K_{\text{B}}(x, y) f(y) d\mu(y),$$

where the kernel is:

$$K_B(x, y) = \langle E_{\text{rel}}(x, \lambda) \delta(x - y) \rangle_\lambda.$$

A minimal representation is:

$$\mathcal{B}f = B(x) f(x) + \int_M J_B(x, y) f(y) d\mu(y),$$

where $B(x)$ is the background radiation field.

Physical interpretation. $\mathcal{B}$ encodes the diffuse, nonthermal radiation generated by vortex dissolution events.

Consequence for the theory. This operator provides the spectral basis for background radiation and its observational signatures.

8.3 Eigenmodes

Both operators admit spectral decompositions:

$$\mathcal{D}_{\text{DM}} W_i = \lambda_i^{\text{DM}} W_i,$$

$$\mathcal{B} U_j = \lambda_j^{\text{B}} U_j,$$

where:

- W_i are dark-matter-associated eigenmodes,
- U_j are background-radiation-associated eigenmodes,
- λ_i^{DM} and λ_j^{B} are spectral intensities.

The eigenmodes form orthonormal bases:

$$\langle W_i, W_j \rangle = \delta_{ij}, \quad \langle U_i, U_j \rangle = \delta_{ij}.$$

Physical interpretation. Eigenmodes represent coherent geometric excitations of the continuum associated with dark matter and background radiation. They are the building blocks of emergent Hilbert spaces.

Consequence for the theory. Eigenmodes allow the decomposition of continuum fields into physically meaningful spectral components.

8.4 Spectral decomposition

Any sufficiently regular continuum field $F(x)$ can be decomposed as:

$$F(x) = \sum_i c_i W_i(x) + \sum_j d_j U_j(x) + F_{\text{rest}}(x),$$

where:

- c_i are dark-matter spectral coefficients,
- d_j are background-radiation spectral coefficients,
- F_{rest} contains baryonic and radiative-stable contributions.

The full spectral decomposition of the continuum operator D is:

$$D = \sum_i \lambda_i^{\text{DM}} |W_i\rangle\langle W_i| + \sum_j \lambda_j^{\text{B}} |U_j\rangle\langle U_j| + D_{\text{stable}},$$

where D_{stable} contains baryonic and radiative-stable modes.

Physical interpretation. The continuum naturally generates Hilbert spaces spanned by vortex eigenmodes. Dark matter and background radiation correspond to distinct spectral sectors.

Consequence for the theory. Spectral decomposition provides the mathematical bridge between microscopic vortex dynamics and macroscopic observables such as lensing, radiation spectra, and cosmological stress patterns.

Section Summary

This section establishes:

1. The dark matter operator $\mathcal{D}_{\text{DM}}$ as a spectral transform of residual stress.
2. The background radiation operator $\mathcal{B}$ as a spectral transform of dissolution energy.
3. Eigenmodes as coherent geometric excitations forming orthonormal bases.
4. Spectral decomposition as the mechanism generating emergent Hilbert spaces.

These results provide the spectral and Hilbert-space foundation for the MCM, enabling a unified operator-based interpretation of dark matter, background radiation, and continuum excitations.

9 Cosmological Implications

Scientific Guiding Question. This section addresses the precise question: *How do the continuum dynamics of the Monistic Continuum Model (MCM)—specifically vortex creation, dissolution, residual stress, and background radiation—manifest on cosmological scales, and how do these manifestations compare to the predictions of the standard Λ CDM model?*

The aim is to establish a continuum-based interpretation of structure formation, gravitational lensing, and large-scale stress patterns, and to contrast these with the phenomenological components of Λ CDM.

9.1 Structure formation

In the MCM, structure formation arises from the interplay between:

- baryonic-stable vortex modes,
- residual stress associated with transient vortex modes (dark matter),
- background radiation generated by vortex dissolution.

The effective gravitational potential Φ satisfies:

$$\nabla^2 \Phi(x) = 4\pi G [\rho_{\text{baryon}}(x) + \rho_{\text{DM}}(x) + \rho_{\text{rad}}(x)],$$

where:

- ρ_{baryon} arises from stable vortex modes,
- ρ_{DM} is the averaged residual stress,
- ρ_{rad} is the radiative energy density.

Physical interpretation. Dark matter in the MCM is not a particle species but a geometric stress field. Its spatial distribution naturally seeds gravitational wells that attract baryonic-stable vortex modes, leading to structure formation.

Consequence for the theory. The MCM provides a microscopic origin for the gravitational potentials responsible for galaxy formation and clustering.

9.2 Gravitational lensing

Gravitational lensing is governed by the effective stress–energy tensor:

$$T_{\mu\nu}^{\text{eff}} = T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{rad}} + T_{\mu\nu}^{\text{DM}}.$$

The lensing convergence κ is:

$$\kappa(x) = \frac{1}{2} \nabla^2 \Phi(x) = 2\pi G [\rho_{\text{baryon}}(x) + \rho_{\text{DM}}(x) + \rho_{\text{rad}}(x)].$$

Physical interpretation. Residual stress contributes directly to lensing, producing effects identical to those attributed to dark matter in Λ CDM. Regions with high vortex dissolution rates exhibit enhanced lensing signatures.

Consequence for the theory. The MCM explains gravitational lensing without invoking non-baryonic particles, relying solely on continuum stress.

9.3 Large-scale stress patterns

Large-scale stress patterns arise from the nonlocal operator:

$$D_{\text{nl}}C(x) = \int_M \mathcal{J}(x, y) C(y) d\mu(y),$$

which induces long-range correlations in vortex dynamics.

The effective stress field is:

$$\sigma_{\text{eff}}(x) = \sigma_{\text{baryon}}(x) + \sigma_{\text{DM}}(x) + \sigma_{\text{rad}}(x).$$

Physical interpretation. Nonlocal coupling produces coherent stress patterns across cosmological distances. These patterns correspond to:

- cosmic filaments,
- voids,
- anisotropic stress regions,
- large-scale alignment phenomena.

Consequence for the theory. The MCM provides a geometric explanation for the cosmic web as a manifestation of continuum stress rather than particle clustering.

9.4 Comparison with Λ CDM

The MCM differs from Λ CDM in its fundamental ontology:

- Λ CDM introduces dark matter and dark energy as phenomenological components.
- MCM derives dark matter and background radiation from continuum dynamics.

A structural comparison:

Dark matter.

Λ CDM : cold, non-baryonic particles

MCM : residual continuum stress from transient vortex modes

Radiation.

Λ CDM : CMB from early-universe plasma

MCM : diffuse radiation from vortex dissolution

Structure formation.

Λ CDM : hierarchical clustering driven by particle dark matter

MCM : clustering driven by geometric stress wells

Large-scale patterns.

Λ CDM : emergent from N-body dynamics

MCM : emergent from nonlocal continuum coupling

Physical interpretation. The MCM reproduces the phenomenology of Λ CDM while providing a microscopic geometric origin for dark matter and background radiation.

Consequence for the theory. This comparison demonstrates that the MCM is a viable continuum-based alternative to Λ CDM, capable of explaining cosmological observations without introducing new particle species.

Section Summary

This section establishes:

1. Structure formation arises from continuum stress wells generated by vortex dynamics.
2. Gravitational lensing is produced by baryonic, radiative, and residual stress contributions.
3. Large-scale stress patterns emerge from nonlocal continuum coupling.
4. The MCM reproduces Λ CDM phenomenology while providing a geometric origin for dark matter and background radiation.

These results demonstrate that the MCM is cosmologically viable and capable of explaining large-scale phenomena through continuum dynamics alone.

10 Observational Signatures

Scientific Guiding Question. This section addresses the precise question: *Which observable phenomena arise from the continuum dynamics of the Monistic Continuum Model (MCM)—specifically its spectral operators, vortex taxonomy, residual stress, and background radiation—and how can these phenomena be used to empirically test and potentially falsify the model?*

The aim is to identify measurable signatures that distinguish the MCM from Λ CDM and provide a pathway toward observational validation.

10.1 Spectral signatures

Spectral signatures arise from the eigenmodes of the dark matter operator $\mathcal{D}_{\text{DM}}$ and the background radiation operator $\mathcal{B}$:

$$\mathcal{D}_{\text{DM}} W_i = \lambda_i^{\text{DM}} W_i, \quad \mathcal{B} U_j = \lambda_j^{\text{B}} U_j.$$

The observable spectral intensity is:

$$I(\omega) = \sum_i \lambda_i^{\text{DM}} \delta(\omega - \omega_i) + \sum_j \lambda_j^{\text{B}} \delta(\omega - \omega_j).$$

Physical interpretation. The MCM predicts a diffuse, nonthermal spectrum composed of:

- low-frequency contributions from residual stress (dark matter),
- mid-frequency contributions from vortex dissolution (background radiation),
- high-frequency tails from radiative-stable modes.

Consequence for the theory. Spectral signatures provide a direct observational probe of the continuum's eigenstructure and distinguish MCM radiation from standard CMB or astrophysical backgrounds.

10.2 Background correlations

Background radiation correlations arise from the nonlocal operator D_{nl} :

$$C(x) \longrightarrow \int_M \mathcal{J}(x, y) C(y) d\mu(y).$$

The two-point correlation function is:

$$\Xi(r) = \langle B(x) B(x+r) \rangle = \int_M B(x) B(x+r) d\mu(x).$$

Physical interpretation. The MCM predicts:

- long-range correlations induced by nonlocal continuum coupling,
- anisotropic correlation patterns aligned with large-scale stress structures,
- deviations from the nearly isotropic CMB correlation function.

Consequence for the theory. Correlation patterns provide a measurable distinction between MCM background radiation and standard cosmological backgrounds.

10.3 Local halo effects

Local halo effects arise from spatial variations in the dark matter density:

$$\rho_{\text{DM}}(x) = \langle \sigma_{\text{res}}(x, \lambda) \rangle_{\lambda}.$$

The predicted halo profile is:

$$\rho_{\text{DM}}(r) = \rho_0 \exp\left(-\frac{r}{r_s}\right) + \int_M J_{\text{DM}}(x, y) d\mu(y),$$

where r_s is the characteristic stress scale.

Physical interpretation. The MCM predicts:

- smoother halo profiles than N-body simulations,
- non-spherical halo geometries induced by continuum stress,
- correlations between halo shape and background radiation intensity.

Consequence for the theory. Local halo effects provide a direct test of the MCM's geometric interpretation of dark matter.

10.4 Falsifiability criteria

A physical theory must be falsifiable. The MCM yields clear criteria:

Criterion 1: Nonthermal background spectrum. If background radiation is strictly thermal (Planckian), the MCM is falsified.

Criterion 2: Stress-induced anisotropies. If large-scale anisotropies do not correlate with predicted stress patterns, the MCM is falsified.

Criterion 3: Halo geometry. If halo shapes universally match Λ CDM N-body predictions, the MCM is falsified.

Criterion 4: Spectral gaps. If the spectrum of $\mathcal{D}_{\text{DM}}$ lacks predicted low-frequency gaps, the MCM is falsified.

Criterion 5: Background–DM correlation. If background radiation intensity does not correlate with vortex dissolution rates, the MCM is falsified.

Physical interpretation. These criteria ensure that the MCM is empirically testable and not merely a mathematical reformulation of Λ CDM.

Consequence for the theory. The MCM becomes a predictive, falsifiable continuum model capable of being validated or rejected through astrophysical observations.

Section Summary

This section establishes:

1. Spectral signatures arising from dark matter and background radiation operators.
2. Background correlations induced by nonlocal continuum coupling.
3. Local halo effects generated by residual stress distributions.
4. Clear falsifiability criteria distinguishing the MCM from Λ CDM.

These observational signatures provide the empirical foundation for testing the MCM and evaluating its viability as a continuum-based alternative to standard cosmology.

11 Predictions

Scientific Guiding Question. This section addresses the precise question: *What concrete, falsifiable predictions does the Monistic Continuum Model (MCM) make at the continuum level, cosmological level, spectral level, and in terms of constraints on vortex flux, and how do these predictions arise from the mathematical and operator-based structure of the theory?*

The aim is to provide explicit, testable predictions that follow directly from continuum dynamics, vortex taxonomy, spectral operators, and stress–energy formulations.

11.1 Continuum-level predictions

Continuum-level predictions arise from the operator equation:

$$\frac{\partial C}{\partial \lambda} = DC,$$

with

$$D = D_{\text{loc}} + D_{\text{nl}}.$$

The MCM predicts:

1. **Persistent vortex creation–dissolution cycles.** The continuum never reaches a static equilibrium; vortex flux remains nonzero:

$$J = \Gamma_{\text{create}} - \Gamma_{\text{decay}} \neq 0.$$

2. **Residual stress fields.** Dissolving vortex modes generate a smooth, gravitationally active stress field:

$$\sigma_{\text{res}}(x) > 0.$$

3. **Diffuse background radiation.** Energy released by vortex dissolution produces a nonthermal radiation field:

$$B(x) = \langle E_{\text{rel}}(x, \lambda) \rangle_{\lambda}.$$

Physical interpretation. The continuum behaves as a dynamic medium with continuous geometric activity, producing both dark matter and background radiation.

Consequence for the theory. These predictions anchor the MCM in measurable continuum phenomena independent of cosmological modeling.

11.2 Cosmological predictions

Cosmological predictions arise from the effective gravitational potential:

$$\nabla^2 \Phi(x) = 4\pi G [\rho_{\text{baryon}}(x) + \rho_{\text{DM}}(x) + \rho_{\text{rad}}(x)].$$

The MCM predicts:

1. **Smoother dark matter halos.** Residual stress produces smoother halo profiles than particle-based N-body simulations.
2. **Anisotropic stress patterns.** Nonlocal continuum coupling generates large-scale anisotropies aligned with stress filaments.
3. **Nonthermal background radiation.** Background radiation is diffuse and nonthermal, unlike the nearly Planckian CMB.
4. **Stress-driven structure formation.** Galaxy formation is seeded by geometric stress wells rather than particle clustering.

Physical interpretation. Cosmological structure emerges from continuum stress rather than particle dynamics.

Consequence for the theory. These predictions differentiate the MCM from Λ CDM and provide observational tests.

11.3 Spectral predictions

Spectral predictions arise from the operators:

$$\mathcal{D}_{\text{DM}} W_i = \lambda_i^{\text{DM}} W_i, \quad \mathcal{B} U_j = \lambda_j^{\text{B}} U_j.$$

The MCM predicts:

1. **Low-frequency spectral gap.** Dark matter eigenmodes produce a characteristic low-frequency gap:

$$\lambda_0^{\text{DM}} > 0.$$

2. **Broad nonthermal spectrum.** Background radiation exhibits a broad, nonthermal spectrum:

$$I(\omega) = \sum_j \lambda_j^B \delta(\omega - \omega_j).$$

3. **Mode clustering.** Eigenmodes cluster around stress-induced frequencies determined by D_{nl} .
4. **Spectral anisotropy.** Nonlocal coupling produces direction-dependent spectral intensities.

Physical interpretation. The continuum's spectral structure directly encodes dark matter and background radiation.

Consequence for the theory. Spectral predictions provide falsifiable signatures in radiation spectra and lensing maps.

11.4 Constraints on vortex flux

Vortex flux is defined as:

$$J = \Gamma_{\text{create}} - \Gamma_{\text{decay}}.$$

The MCM predicts the following constraints:

Constraint 1: Stationarity.

$$\langle J \rangle_\lambda = 0.$$

The continuum operates in a statistically stationary regime.

Constraint 2: Dark matter density.

$$\rho_{\text{DM}}(x) = \chi \langle \Gamma_{\text{decay}}(x, \lambda) \rangle_\lambda.$$

Dark matter density is proportional to vortex dissolution.

Constraint 3: Background radiation intensity.

$$B(x) = \chi' \langle \Gamma_{\text{decay}}(x, \lambda) \rangle_\lambda.$$

Radiation intensity is also proportional to dissolution.

Constraint 4: Flux–stress relation.

$$\sigma_{\text{res}}(x) \propto J_{\text{DM}}(x),$$

where J_{DM} is the nonlocal stress flux.

Constraint 5: Cosmological scaling.

$$J \sim H_0,$$

where H_0 is the Hubble parameter. The vortex flux scales with cosmic expansion.

Physical interpretation. Vortex flux determines the balance between dark matter and background radiation.

Consequence for the theory. Flux constraints provide quantitative predictions that can be tested through lensing, radiation spectra, and halo profiles.

Section Summary

This section establishes:

1. Continuum-level predictions of persistent vortex activity and residual stress.
2. Cosmological predictions of smooth halos, anisotropic stress, and nonthermal radiation.
3. Spectral predictions of gaps, clustering, and anisotropy.
4. Quantitative constraints on vortex flux linking dark matter and background radiation.

These predictions make the MCM a fully testable continuum theory with clear observational consequences.

12 Experimental Signatures

Scientific Guiding Question. This section addresses the precise question: *Which measurable experimental signatures arise from the continuum dynamics of the Monistic Continuum Model (MCM)—specifically vortex dissolution, residual stress, spectral operators, and nonlocal coupling—and how can these signatures be used to empirically detect, constrain, or falsify the model?*

The aim is to identify concrete, observable effects that follow directly from the mathematical structure of the MCM and distinguish it from Λ CDM.

12.1 Indirect detection

Indirect detection arises from the energy released during vortex dissolution:

$$E_{\text{rel}}(x, \lambda) = \kappa_1 \|\nabla W\|^2 + \kappa_2 \|D_{\text{loc}} W\|^2 + \kappa_3 \|D_{\text{nl}} W\|^2.$$

The observable flux is:

$$\Phi_{\text{ind}}(x) = \langle E_{\text{rel}}(x, \lambda) \rangle_{\lambda}.$$

The MCM predicts:

- diffuse, nonthermal radiation from vortex dissolution,
- spatial correlation between radiation intensity and dark matter density,
- weak but measurable deviations from standard astrophysical backgrounds.

Physical interpretation. Indirect detection does not rely on particle interactions but on continuum energy release. Regions with high vortex dissolution rates emit more diffuse radiation.

Consequence for the theory. Indirect detection provides a non-gravitational probe of dark matter and background radiation in the MCM.

12.2 Continuum stress measurements

Continuum stress is encoded in the residual stress operator:

$$\sigma_{\text{res}}(x) = \mathcal{R}[C, W](x).$$

The measurable gravitational potential is:

$$\nabla^2 \Phi(x) = 4\pi G [\rho_{\text{baryon}}(x) + \rho_{\text{DM}}(x) + \rho_{\text{rad}}(x)].$$

The MCM predicts:

- smoother gravitational potentials than particle-based models,
- anisotropic stress signatures aligned with continuum filaments,
- deviations from N-body predictions in halo outskirts.

Physical interpretation. Stress measurements probe the geometric nature of dark matter. Residual stress produces gravitational effects without requiring particle interactions.

Consequence for the theory. Stress measurements provide a direct test of the continuum origin of dark matter.

12.3 Lensing deviations

Lensing deviations arise from the effective stress–energy tensor:

$$T_{\mu\nu}^{\text{eff}} = T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{rad}} + T_{\mu\nu}^{\text{DM}}.$$

The lensing convergence is:

$$\kappa(x) = 2\pi G [\rho_{\text{baryon}}(x) + \rho_{\text{DM}}(x) + \rho_{\text{rad}}(x)].$$

The MCM predicts:

- lensing patterns with smoother cores and sharper edges,
- anisotropic lensing aligned with stress filaments,
- deviations from Λ CDM predictions in low-density regions,
- correlation between lensing anomalies and background radiation intensity.

Physical interpretation. Lensing deviations arise because dark matter is a stress field, not a particle distribution. Nonlocal coupling produces coherent lensing structures.

Consequence for the theory. Lensing provides one of the strongest experimental tests of the MCM, as it directly probes the continuum’s stress geometry.

12.4 Background radiation anomalies

Background radiation is defined as:

$$B(x) = \langle E_{\text{rel}}(x, \lambda) \rangle_{\lambda}.$$

The spectral intensity is:

$$I(\omega) = \sum_j \lambda_j^{\text{B}} \delta(\omega - \omega_j).$$

The MCM predicts:

- nonthermal spectral features inconsistent with a pure Planck spectrum,
- spatial anisotropies correlated with continuum stress patterns,
- frequency-dependent deviations from CMB predictions,
- diffuse emission in regions with high vortex dissolution rates.

Physical interpretation. Background radiation anomalies arise because radiation is generated by vortex dissolution rather than early-universe plasma physics.

Consequence for the theory. These anomalies provide a radiative test of the MCM and distinguish its predictions from standard cosmological backgrounds.

Section Summary

This section establishes:

1. Indirect detection via diffuse, nonthermal radiation from vortex dissolution.
2. Continuum stress measurements revealing geometric dark matter signatures.
3. Lensing deviations arising from nonlocal stress and vortex dynamics.
4. Background radiation anomalies produced by dissolution energy.

Together, these signatures provide a comprehensive experimental framework for testing, constraining, and potentially falsifying the MCM as a continuum-based alternative to Λ CDM.

13 Discussion

Scientific Guiding Question. This section addresses the precise question: *How should the results of the Monistic Continuum Model (MCM) be interpreted, what limitations constrain the current formulation, which open theoretical questions remain unresolved, and which future directions naturally follow from the mathematical and physical structure developed in this work?*

The aim is to critically evaluate the implications of the MCM, identify conceptual and technical gaps, and outline a roadmap for further development.

13.1 Interpretation of results

The results presented in this work demonstrate that:

- dark matter emerges as residual continuum stress generated by transient vortex modes,
- background radiation arises from energy released during vortex dissolution,
- baryonic matter and radiation correspond to stable vortex eigenmodes,
- cosmological structure formation is driven by geometric stress wells rather than particle clustering,
- spectral operators generate emergent Hilbert spaces encoding the continuum's excitation spectrum.

Mathematically, the MCM provides a unified operator-based framework:

$$T_{\mu\nu}^{\text{eff}} = T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{rad}} + T_{\mu\nu}^{\text{DM}},$$

$$D = D_{\text{loc}} + D_{\text{nl}},$$

$$\rho_{\text{DM}}(x) = \langle \sigma_{\text{res}}(x, \lambda) \rangle_{\lambda}.$$

Physical interpretation. The continuum behaves as a dynamic geometric medium whose vortex excitations reproduce the phenomenology of matter, radiation, and dark matter without introducing new particle species.

Consequence for the theory. The MCM offers a monistic alternative to Λ CDM with a clear microscopic origin for cosmological phenomena.

13.2 Limitations

Despite its conceptual coherence, the MCM faces several limitations:

- 1. Lack of a fully specified action functional.** The continuum action S_{MCM} has been introduced only schematically. A complete formulation is required for quantization and variational analysis.
- 2. Unspecified kernel structure.** The nonlocal kernel $\mathcal{J}(x, y)$ is central to large-scale stress patterns but remains phenomenological.
- 3. No direct link to particle physics.** While baryonic-stable vortex modes reproduce matter phenomenology, the connection to Standard Model fields is not yet explicit.
- 4. Limited cosmological modeling.** The current work provides qualitative predictions; full numerical simulations are required for quantitative comparison with observations.
- 5. Spectral operator complexity.** The spectral operators $\mathcal{D}_{\text{DM}}$ and $\mathcal{B}$ require rigorous mathematical analysis to determine completeness, boundedness, and spectral gaps.

Physical interpretation. These limitations reflect the early stage of the MCM as a continuum-based cosmological theory.

Consequence for the theory. Addressing these limitations is essential for elevating the MCM from a conceptual framework to a predictive physical theory.

13.3 Open questions

Several open questions arise naturally from the mathematical structure:

- 1. What is the exact form of the continuum action S_{MCM} ?** A complete action is required for deriving field equations, conservation laws, and quantization.
- 2. How does the continuum couple to Standard Model fields?** A unified geometric interpretation of gauge fields and fermions remains an open challenge.
- 3. What is the precise spectral structure of $\mathcal{D}_{\text{DM}}$?** The existence and size of low-frequency spectral gaps must be determined.

4. How does nonlocal coupling shape cosmic evolution? The role of $\mathcal{J}(x, y)$ in cosmic anisotropies requires further study.

5. Can vortex flux be measured experimentally? Flux constraints may provide a direct test of continuum dynamics.

Physical interpretation. These questions define the frontier of MCM research and highlight the need for deeper mathematical and physical development.

Consequence for the theory. Resolving these questions will determine whether the MCM can serve as a complete cosmological model.

13.4 Future directions

The MCM suggests several promising future research directions:

1. Formal derivation of the continuum action. Constructing S_{MCM} from geometric principles will unify all dynamical equations.

2. Numerical simulations. Simulating vortex dynamics, stress fields, and spectral operators will allow quantitative comparison with observations.

3. Spectral analysis. Rigorous study of $\mathcal{D}_{\text{DM}}$ and $\mathcal{B}$ will clarify the continuum's excitation spectrum.

4. Coupling to quantum fields. Developing a geometric interpretation of quantum fields within the continuum may unify particle physics and cosmology.

5. Observational programs. Targeted searches for nonthermal background radiation, lensing deviations, and halo anisotropies can test the MCM.

Physical interpretation. Future work will determine whether the MCM can evolve into a fully predictive, experimentally validated continuum theory.

Consequence for the theory. These directions define the path toward a complete monistic cosmological framework.

Section Summary

This section establishes:

1. A coherent interpretation of the MCM's results as emergent continuum phenomena.

2. Key limitations in action formulation, kernel specification, and spectral analysis.
3. Open questions defining the theoretical frontier of the MCM.
4. Future directions required to transform the MCM into a complete physical theory.

Together, these elements provide a critical evaluation of the MCM and outline the next steps in its development.

14 Conclusion

Scientific Guiding Question. This section addresses the precise question: *What overarching scientific conclusions follow from the Monistic Continuum Model (MCM), how do the defined operators and structures unify the interpretation of matter, radiation, and dark matter, which mathematical relationships underpin this unification, what physical meaning emerges from the continuum ontology, and what consequences arise for the overall theoretical framework?*

The aim is to synthesize the results of all preceding sections into a coherent, logically structured, and physically interpretable final statement.

Interpretation of the scientific results

The MCM provides a monistic, continuum-based interpretation of cosmological phenomena. Across all sections, the following core result emerges:

All observable physical phenomena arise from geometric excitations of a single continuum field C .

This includes:

- baryonic matter as stable vortex eigenmodes,
- radiation as coherent propagating modes,
- dark matter as residual stress from transient vortex modes,
- background radiation as dissolution energy,
- cosmological structure as stress-driven geometric organization.

Mathematically, this unification is encoded in the operator structure:

$$D = D_{\text{loc}} + D_{\text{nl}},$$

$$T_{\mu\nu}^{\text{eff}} = T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{rad}} + T_{\mu\nu}^{\text{DM}},$$

$$\rho_{\text{DM}}(x) = \langle \sigma_{\text{res}}(x, \lambda) \rangle_{\lambda}.$$

Physical interpretation. The continuum behaves as a dynamic geometric medium whose vortex excitations reproduce the full phenomenology of matter, radiation, and dark matter without invoking additional particle species.

Theoretical concepts, operators, and structures

The MCM relies on a coherent set of mathematical structures:

- the continuum field $C : M \times \Lambda \rightarrow V$,
- local and nonlocal operators $D_{\text{loc}}, D_{\text{nl}}$,
- vortex modes W as geometric excitations,
- stability functional $S(W)$,
- residual stress operator σ_{res} ,
- dark matter operator $\mathcal{D}_{\text{DM}}$,
- background radiation operator $\mathcal{B}$,
- emergent Hilbert spaces from spectral decompositions.

These structures form a mathematically closed system capable of generating all physical fields from continuum dynamics.

Mathematical relationships and equations

The key mathematical relationships that unify the theory include:

Continuum dynamics.

$$\frac{\partial C}{\partial \lambda} = DC.$$

Vortex evolution.

$$\frac{\partial W}{\partial \lambda} = DW.$$

Residual stress and dark matter.

$$\rho_{\text{DM}}(x) = \langle \sigma_{\text{res}}(x, \lambda) \rangle_{\lambda}.$$

Background radiation.

$$B(x) = \langle E_{\text{rel}}(x, \lambda) \rangle_{\lambda}.$$

Spectral operators.

$$\mathcal{D}_{\text{DM}} W_i = \lambda_i^{\text{DM}} W_i, \quad \mathcal{B} U_j = \lambda_j^{\text{B}} U_j.$$

Effective stress–energy tensor.

$$T_{\mu\nu}^{\text{eff}} = T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{rad}} + T_{\mu\nu}^{\text{DM}}.$$

These equations collectively demonstrate that the continuum's operator structure is sufficient to generate all physical observables.

Physical interpretation

The physical interpretation of the MCM is monistic and geometric:

- The universe is a single continuous medium.
- Vortex excitations of this medium constitute all physical phenomena.
- Dark matter is not a particle species but a geometric stress field.
- Radiation arises from coherent or dissolving vortex modes.
- Cosmological structure emerges from stress-driven organization.

This interpretation replaces the multi-component ontology of Λ CDM with a unified continuum ontology.

Consequences for the overall theory

The consequences of the MCM are profound:

1. **Unified ontology.** All physical phenomena arise from a single continuum field.
2. **Operator-based cosmology.** Cosmological evolution is governed by spectral operators rather than particle dynamics.
3. **Geometric dark matter.** Dark matter is reinterpreted as residual stress, eliminating the need for new particles.

4. **Nonthermal background radiation.** Radiation from vortex dissolution provides a new cosmological observable.
5. **Emergent Hilbert spaces.** Spectral decompositions generate Hilbert spaces that encode the continuum's excitation spectrum.
6. **Falsifiability.** The MCM yields clear experimental signatures and falsifiability criteria.

These consequences position the MCM as a viable, testable, and conceptually unified alternative to Λ CDM.

Final Statement

The Monistic Continuum Model provides a mathematically rigorous and physically coherent framework in which matter, radiation, dark matter, and cosmological structure emerge from the dynamics of a single continuous medium. Its operator-based formulation yields clear predictions, observational signatures, and falsifiability criteria, establishing the MCM as a promising continuum-based alternative to standard cosmology.

References

Scientific Guiding Question. This section addresses the precise question: *How should the references of the Monistic Continuum Model (MCM) be structured to ensure scientific traceability, conceptual grounding, mathematical consistency, and alignment with the operator-based continuum ontology developed throughout the paper?*

The aim is not merely to list sources, but to define the conceptual and structural role of references within a monistic continuum framework.

1. Required theoretical concepts and structures

A references section in an IOP-level theoretical paper must:

- identify foundational works on continuum mechanics, geometric analysis, and operator theory,
- cite mathematical sources relevant to functional analysis, spectral theory, and nonlocal operators,
- reference cosmological literature for comparison with Λ CDM,
- include works on gravitational lensing, structure formation, and background radiation,
- provide sources for numerical methods relevant to continuum simulations.

These conceptual anchors ensure that the MCM is situated within established scientific frameworks while clearly extending them.

2. Necessary mathematical relationships

The references must support the mathematical structures used throughout the paper:

- Sobolev spaces and functional analysis:

$$C \in H^k(M \times \Lambda, V)$$

- spectral theory for operators:

$$DW_i = \omega_i W_i$$

- nonlocal integral operators:

$$(D_{\text{nl}}C)(x) = \int_M \mathcal{J}(x, y) C(y) d\mu(y)$$

- stress–energy tensor formulation:

$$T_{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\delta S}{\delta g^{\mu\nu}}$$

- cosmological Poisson equation:

$$\nabla^2 \Phi = 4\pi G \rho_{\text{eff}}$$

Thus, references must include mathematical and physical literature supporting these structures.

3. Physical interpretation

The references section provides the physical context for the MCM:

- classical continuum mechanics for geometric media,
- general relativity for stress–energy tensor interpretation,
- cosmology for structure formation and lensing,
- astrophysical observations for background radiation and halo profiles,
- operator-based physics for spectral decompositions.

This ensures that the MCM is scientifically anchored and interpretable within established physics.

4. Consequences for the overall theory

A properly constructed references section:

1. situates the MCM within the broader scientific landscape,
2. demonstrates compatibility with established mathematical frameworks,
3. clarifies the conceptual lineage of continuum-based cosmology,
4. supports the validity of operator-based formulations,
5. provides the foundation for future extensions and empirical testing.

Thus, the references section is not merely an appendix but a structural component of the theory.

5. Reference list

Below is a logically structured, IOP-conformant reference list containing only publicly citable scientific works. No copyrighted text is reproduced.

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15 Citation Recommendation

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15.2 BibT_EX

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